Contents

1. Introduction

Carbon capture plays a crucial role in reducing the concentration of carbon dioxide released into the atmosphere by various industries. One significant contributor to carbon dioxide emissions is the hydrogen production industry, particularly through the Steam Methane Reforming (SMR) process. SMR is a conventional method widely used not only for hydrogen production but also in applications such as ammonia synthesis, oil refining, methanol production, chemical manufacturing, fuel cell systems, and other emerging low-carbon energy technologies. This process requires high temperatures to convert methane into hydrogen and carbon dioxide, making it a key focus area for carbon capture efforts aimed at mitigating environmental impact [1].

$$\text{Steam Methane Reforming reaction: } CH_4 + H_2O \rightarrow CO + 3H_2$$

The products of the reaction are further processed to produce carbon dioxide and hydrogen

$$\text{Water-Gas Shift (WGS) reaction: } CO + H_2O \rightarrow CO_2 + H_2$$

The carbon dioxide produced during the SMR process is captured before being released into the atmosphere, thereby reducing the carbon content in the air. Separated carbon dioxide, extracted from process gas or flue gas, can be dried, compressed, and prepared for storage or utilization. Therefore, this study focuses on assessing the environmental performance of CO_2 capture in an SMR-based hydrogen production process [2].

2. Goal and Scope

The purpose of this study is to analyze the life cycle assessment (LCA) of the carbon capture system integrated with the Steam Methane Reforming (SMR) process. Specifically, the study aims to evaluate the capture of 1 kg of CO_2 and to assess the effectiveness of this carbon capture system in reducing carbon emissions from SMR-based hydrogen production [3].

This study aims to:

Quantify the material and energy inputs required to capture 1 kg of CO_2 [5, 7]. Identify key areas of environmental impact within the carbon capture system. Evaluate potential improvements in CO_2 separation processes, solvent or sorbent circulation, regeneration methods, heat supply, electricity usage, and CO_2 compression. Facilitate better decision-making to effectively reduce emissions from traditional SMR hydrogen production. The results will provide a foundational understanding of the environmental performance of CO_2 capture in SMR and serve as a reference for comparison with other carbon capture or emission reduction technologies.

3. Functional Unit

The functional unit is defined as **1 kg of concentrated CO_2 obtained through the Steam Methane Reforming (SMR) method** [5, 7]. The analysis serves as a foundation for assessing all of the energy and material inputs needed to capture one kilogram of CO_2 produced during SMR. It is intended to make direct study of the SMR process with the carbon capture step and to comprehend the decrease in the rate of CO_2 emission rate attributable to the integration of the capture unit [6].

This functional unit is consistent with those adopted in comparable LCA studies of post-combustion carbon capture systems and allows direct benchmarking against the well-documented emissions profile of conventional SMR [5, 8].

4. System Boundary

The boundary is based on the **Cradle-to-Gate** method [8]. The natural gas (CH_4) intake to the SMR plant is the moment the cradle part begins. It includes the material and energy balance of CH_4 and water with hydrogen gas generation, CO_2, CO slip, and any other energy inputs related to it [4, 5]. The primary focus is on CO_2 absorption and separation from the primary process stream, solvent regeneration, and the energy input required for capturing the carbon, as well as the compression of CO_2 up to the gate in contrast to the production of hydrogen [7].

The following are excluded from the system boundary [6, 8].

Infrastructure according to plant design and equipment decommissioning at the end of its useful life are not considered. The transportation, storage and use of the captured carbon are also excluded. The use of co-produced hydrogen gas at the end falls outside the defined boundary.

5. References

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